

ANIRUDHA SUTAR

AI Agent Engineer · Generative AI · Agentic AI · LLM Applications · RAG · LangGraph · Python

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Professional Summary

AI Agent Engineer with 2.5+ years of experience building Generative AI and LLM applications, agentic workflows, and Retrieval-Augmented Generation (RAG) pipelines for insurance and enterprise use cases. Hands-on with LangGraph and LangChain agent orchestration, tool calling, conversation memory, vector databases, and graph analytics; ships Python and FastAPI services with LLM observability, grounded generation, and explainable AI on Docker and AWS.

Technical Skills

AI Agents & LLM	LangGraph, LangChain, AI Agents, Agentic Workflows, Multi-Agent Orchestration, Tool / Function Calling, MCP (Model Context Protocol), Prompt Engineering, Structured Outputs, Conversation Memory, Groq, Ollama
RAG & Retrieval	Retrieval-Augmented Generation (RAG), Agentic RAG, Corrective RAG, Hybrid Search (BM25 + Semantic), Query Rewriting, Embeddings, Vector Databases (Pinecone, OpenSearch, ChromaDB, FAISS), Sentence-Transformers, Tavily
LLMOps & Quality	LangSmith Tracing, LLM Observability, Token Usage and Cost Optimization, Response Caching, Evidence Grading, Hallucination Mitigation, Audit Logging
Backend & Data	Python, FastAPI, REST APIs, Pydantic, Streamlit, Redis, SQL, MySQL, SQLite, Neo4j, Neo4j Graph Data Science (GDS), Graph Analytics, Document Intelligence
Cloud & DevOps	AWS (EC2, S3, Lambda), Docker, Git, GitHub Actions, CI/CD, Claude Code
ML & Computer Vision	PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NLP, OpenCV, CNN, Transfer Learning, BERT

Professional Experience

Associate Software Engineer (AI) · Insuremo Pvt Ltd

Apr 2025 – Present

MMV Mapping System for Motor Insurance

LangGraph · RAG · Hybrid Search (BM25 + Semantic) · OpenSearch · Redis · LangSmith · Prompt Engineering

- Built a scalable LangGraph service to standardize and map vehicle data (Make-Model-Variant) across insurers with inconsistent formats, reducing manual mapping effort by an estimated 40%.
- Implemented hybrid retrieval combining BM25 keyword search with semantic vector search over an OpenSearch vector database to surface more accurate candidates, grounding LLM outputs in verified vehicle records through a RAG layer.
- Cached match responses in Redis so repeated queries are served from the cache instead of re-running retrieval and LLM calls, lowering latency and token cost.
- Monitored outputs and token usage in LangSmith to tune the workflow; selected a fast, cost-efficient LLM and compacted prompts to cut context size without losing match accuracy, with configurable strict/relaxed thresholds.

Fraud & Collusion Detection in Claims

Neo4j GDS · LangGraph · Python · AI Agents

- Modeled claims, claimants, garages, and repair entities as a knowledge graph in Neo4j; applied WCC and Leiden community detection to surface collusion networks across 10K+ synthetic claim records.
- Layered multi-signal fraud detection (narrative-to-damage mismatches, overcharged parts, invoice inconsistencies) on top of the graph, cutting false-positive manual reviews by an estimated 30%.
- Orchestrated the end-to-end pipeline with LangGraph-based AI agents that generate natural-language explanations, making flagged claims interpretable for non-technical reviewers.

Policy Document Template Generation System

AI Agents · Aspose · Document Parsing · Dynamic Tagging

- Built an AI-agent-driven document intelligence pipeline converting static insurance policy documents into dynamic, tag-based templates with Aspose, cutting manual policy-generation effort by an estimated 30%.
- Implemented dynamic tagging to auto-populate templates from structured policy objects, using prompt engineering to extract structured fields from unstructured source documents.

- Developed computer vision applications for the agriculture sector using TensorFlow, scikit-learn, and transfer learning on pretrained VGG and Xception models.
- Delivered models to users through a FastAPI inference backend and a React frontend, streamlining the interface and data processing.
- Collaborated on designing and training machine learning models, optimizing accuracy and performance for agricultural applications.

Projects

SunDesk Copilot – Agentic RAG Support Assistant

2026

LangGraph · Groq · Pinecone · Tavily · FastAPI · Streamlit · LangSmith · Docker | github.com/anirudh6370/Sundesck-Copilot

- Built a corrective, agentic RAG assistant as a stateful LangGraph workflow that routes each query, retrieves from a Pinecone vector database, uses the LLM to grade evidence, falls back to Tavily web search, and rewrites and retries weak queries.
- Reduced hallucination risk by generating only from graded evidence with source citations, labelling web-sourced answers, and returning an explicit insufficient-evidence response instead of guessing.
- Implemented multi-turn conversation memory with per-thread SQLite checkpointing and history-aware query rewriting, so follow-up questions resolve against earlier turns.
- Enforced JSON-schema structured outputs (Pydantic) for routing and grading, and added LangSmith tracing plus a SQLite audit log for end-to-end LLM observability.
- Shipped a document ingestion pipeline (PDF, DOCX, Markdown; chunking with local sentence-transformers embeddings), a FastAPI REST API, and a Streamlit chat UI.

Alzheimer's Early Diagnosis with Brain MRI Processing

Dec 2023

OpenCV · VGG16 · CNN | [github · Alzheimer-MRI-detection](https://github.com/Alzheimer-MRI-detection)

- Built an early Alzheimer's detection model on brain MRI images using VGG16 feature extraction and CNN classification.
- Applied Gaussian blurring and skull stripping in preprocessing, balanced the classes, and used data augmentation to improve early-stage detection.

Education & Certifications

M.Sc. in Data Science · Silicon University, Bhubaneswar

CGPA: 9.33 / 10 · 2022 – 2024

Certifications & Achievements:

- TensorFlow Python – Great Learning
- Python for Data Analysis – Great Learning
- Achievers Club Member – Silicon University